

KardiaChain: Bringing blockchain to the masses in South East Asia with the cloud

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ABOUT KARDIACHAIN

With Google Cloud delivering the required availability, security, cost-effectiveness, blockchain offerings, KardiaChain can save time and resource in product development, increasing blockchain adoption.



About KardiaChain

"Founded in 2018, KardiaChain provides hybrid blockchain infrastructure and solutions for businesses and government organizations in South East Asia. Kardiachain's flagship innovation is Kyokai, a blockchain-based bridging solution to transcend the boundaries between physical and digital world. The organization has 150 employees working from offices in Vietnam, Singapore, and Taiwan, and has grown to over 100 partners and 700,000 public users."

Industries: Technology

Google Cloud results

- Reduces monthly infrastructure costs by 27%
- Accelerates onboarding of new customers by 15x
- Cuts database migration downtime more than 4x
- Enables the business to build the next iteration of a product designed to accelerate blockchain adoption at scale

Location: Vietnam

Deliver 100% availability, optimize customer partner experience the blockchain

Cloud Load Balancing

Google Cloud (<https://cloud.google.com/>),

Kubernetes Engine

Google Cloud Armor
(<https://cloud.google.com/armor/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

According to forecasts

(<https://www.pwc.com/gx/en/industries/technology/publications/blockchain-report-transform-business-economy.html>)

, blockchain technology will disrupt businesses and

governments, affecting 40 million jobs worldwide.

Analysts International Data Corp. describes

(<https://blogs.idc.com/2020/12/14/how-blockchain-plays-into-digital-transformation/>)

blockchains as digital, distributed ledgers of transactions and records that increase cross-border transaction performance, data resiliency and security, while lowering the cost of doing business. Unsurprisingly, blockchain is attracting investment, innovation and development from businesses in established and emerging economies.

Founded in 2018 with the team based in Southeast Asia and Silicon Valley, KardiaChain

(<https://kardiachain.io/>) aims to 'bring blockchain to the masses' with a blockchain network that connects other networks and applications in the blockchain ecosystem.

"We see a very active hub with a young population in Vietnam, so we elected to create solutions that were easy and inexpensive to use thanks to low transaction fees," explains Huy Nguyen, Co-founder, KardiaChain. "With Google Cloud providing a scalable, dynamic cloud infrastructure, we plan to continue our development activities and promote blockchain and web3 as the pillars of next-generation technologies."

The ultimate goal, according to its founders, is for KardiaChain to create a unified ecosystem where developers can easily create smart contracts that can run on multiple blockchains. This helps optimise costs and avoid congestion as smart contracts can connect to other chains in a trustless and secure

manner. The blockchain operator runs what it describes as a dual node system to enable inter-chain operations among existing and upcoming blockchain platforms.

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—Huy Nguyen, Co-founder, KardiaChain

Strategic alignment with Google Cloud to drive migration

As a startup, KardiaChain had focused on managing costs and started operations on a traditional cloud provider that provided startup credits. However, as the business and its blockchain products and services matured, its leadership decided to migrate the KardiaChain blockchain to Google Cloud (<https://cloud.google.com/>).

"Quite a few engineers on our team were already familiar with Google Cloud products and services, which made setup on the new platform very easy," explains Huy. "We also felt that the flexible infrastructure of Google Cloud aligned well from a strategic and ecosystem development perspective."

Working with Google Cloud teams and a local partner, the business completed its move to Google Cloud in late 2022. Cloud migration tools enabled a smooth transition from the previous cloud environment to Google Cloud, and surprised Huy with the minimal disruption involved. Using Database Migration Service (<https://cloud.google.com/database-migration>), the business was able to reduce migration downtime more than fourfold, from a predicted 24 hours to just five hours, while the partner assisted with the seamless migration of virtual machines across to Google Cloud.

Achieving the near-100% availability needed for public blockchain

Huy Nguyen describes availability and security as "the most important criteria in the blockchain

infrastructure world," and Google Cloud is helping the business fulfil its objectives in both areas.

"Because we are running our blockchain products in a scalable, resilient environment with Compute Engine (<https://cloud.google.com/compute>), we are achieving the near-100% uptime needed in a public blockchain environment," he says. "Our confidence in Google Cloud is such that we are inviting the parties who validate blockchain transactions, and its partners, to use the cloud service."

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founder, KardiaChain*

The business is also taking advantage of Google Kubernetes Engine autopilot mode, in which Google manages its cluster configuration, including nodes, scaling and security, to focus its resources on product development.

"With Google Kubernetes Engine in place, we can focus on building a blockchain built-in solution that enables builders and users to drive blockchain adoption at scale," says Huy. KardiaChain also uses

Cloud Load Balancing

(<https://cloud.google.com/load-balancing>) and **Cloud Armor** (<https://cloud.google.com/armor>) to mitigate security risks such as distributed denial of service (DDoS) attacks designed to overload targeted digital properties with traffic.

Cost optimization is also important to a business with the ambitions of KardiaChain. Monthly infrastructure bills have fallen by 27% with Google Cloud—freeing budget to focus on projects such as development of Kyokai, the next iteration of its blockchain ecosystem. "With a cost-effective infrastructure now in place, we can focus on building a blockchain built-in solution that enables builders and users to drive blockchain adoption at scale," says Huy.

Onboarding new customers to its blockchain network quickly and efficiently is also key to KardiaChain's growth plans, and the business has made this process a full 15 times faster with Google Cloud than with its previous provider.



Building data analytics capabilities

With Google Cloud providing a resilient, secure and scalable infrastructure to support development of the KardiaChain ecosystem, the business is now

looking to the cloud provider to build up its data analytics capability. "We're looking at BigQuery and the product-set related to data warehousing to leverage the public data captured in blockchain and related data," says Huy. "Collecting and analyzing that data could deliver a lot of value to us."

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KardiaChain is now well positioned itself as a key enabler in South East Asia of convergence between the online and offline worlds. "The physical world is being mapped to the digital world and we are working with the Vietnam Government to incorporate the metaverse into its digital transformation strategy," says Huy. "This involves bringing documents, identities and locations into the digital environment, and the blockchain, with the support of Google Cloud technologies, enables this to happen faster and more efficiently."

